



Short Questions:

1. What is R programming language commonly used for? How does R differ from other programming languages?
2. When was R programming language first developed? Who are the key contributors to the development of R?
3. What steps are involved in installing R on a Windows system? Are there alternative methods to install R on a macOS?
4. What is the purpose of an if-else statement in R? How is a switch statement different from an if-else statement?
5. Define a function in R that adds two numbers.
6. You have a dataset containing information about temperatures. What data type would you use for this variable, and why?
7. How is a user-defined function created in R?
8. You need to perform a task repeatedly until a certain condition is met. How would you implement a while loop for this task?
9. You have a variable num with the value 17. How would you use an operator to check if num is an even number?
10. How is a matrix different from a data frame in R?

Essay Questions:

1. Describe the fundamental concepts of R programming, including its history, key features, and basic syntax. Explain how R differs from other programming languages and provide examples of its applications in data analysis and visualization.
2. Explore the role of variables, data types, and data structures in R programming. Discuss how these elements contribute to effective data manipulation and analysis, and provide real-world examples of scenarios where choosing the appropriate data structures is crucial.
3. Examine the significance of control structures such as loops and decision-making statements in R programming. Discuss how these structures enhance the efficiency and flexibility of code.
4. Investigate the advantages and disadvantages of using R for statistical analysis. Provide examples of statistical functions and methods available in R, and discuss how R facilitates exploratory data analysis.



5. Explore the role of R in data visualization. Discuss the various plotting functions available in R, such as ggplot2, and explain how they contribute to creating informative and visually appealing graphs.

Case Study Question:

Analyzing Customer Behavior with R

You are a data analyst for a retail company that sells products both in physical stores and online. The company is interested in understanding customer behavior to optimize its marketing strategies. You have been provided with a dataset containing information about customer purchases, including the products bought, the purchase amounts, and whether the purchase was made in-store or online. Using R programming, design an analysis plan to extract valuable insights

- Perform exploratory data analysis to understand the structure of the data (assume your own data).
- Identify key variables and their distributions.
- Use R programming language to identify the data types of variables involved.